

Szymon Sarnowicz

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Experience

General Motors – Software Engineer II | *C, C++, Python* | *Feb 2023 to present*

- Designed fault detection pipeline in C/C++ with distributed monitoring and API integrations to prevent ~\$480M in cascading system failures; maintained production stability across complex customer workflows.
- Worked directly with cross-functional stakeholders (firmware, systems, product) to translate complex requirements into robust technical architectures and deployments.
- Implemented end-to-end CI/CD integration and automated system-level testing framework in Python; established monitoring and feedback loops for rapid iteration and incident prevention.

Data4Good – University of Michigan – Lead Developer, Hangul Project | *Python* | *Dec 2023 to Dec 2024*

- Embedded with UN customers to understand document processing workflows and deployed tailored AI infrastructure translating their requirements into immediate, measurable impact at scale.
- Designed application to use Apache Tika for HTML parsing and BART model for NLP-based extraction, redesigning the API pipeline into stages with exponential backoff to increase success rate from 53% to 92%.

Readiscover AI – Technical Lead | *Python, JavaScript* | *Jan 2026 to present*

- Built full-stack semantic search platform for long-form text retrieval with adaptive chunking and multi-layer caching (GCS for persistent storage, Redis for sub-second cache access with optimized TTL).
- Implemented LLM query reformulation (Gemini 2.5 Flash at temperature 0.0 for deterministic output) before embedding, prioritizing trustworthiness by returning only continuous verbatim passages and eliminating hallucination risk (vs. 43.6% for GPT-5.1).
- Operated independently across full stack: infrastructure, integrations, and customer-facing performance metrics.
- Conducted rigorous ablation testing across chunking parameters and evaluated against GPT-5.1 on 50 test questions achieving 34% top-3 accuracy with zero hallucinations and superior proximity to correct passages, with results published at University of Michigan and deployed as production platform.

General Motors – Program Manager, Autonomous Systems (Ultra Cruise) | *Jul 2021 to Feb 2023*

- Tripled data collection per sprint by automating data-request workflows and authored diagnostic frameworks enabling cross-functional teams to independently troubleshoot sensor integration issues, accelerating iteration cycles and reducing development time through infrastructure and customer enablement.

General Motors – Software Engineer (Rotational Program) | *C, C++, Python* | *Sep 2019 to Jul 2021*

- Automated calibration data management processes across embedded systems, eliminating 1 hour of daily manual data entry while gaining foundational experience in embedded systems, real-time data handling, sensor calibration, and automotive software development practices.

Education

Master's in Applied Data Science, University of Michigan, summa cum laude | *Jan 2023 to Dec 2025*

B.S. Mechanical Engineering, University of Washington, cum laude | *Sep 2015 to Jun 2019*

Technical Skills & Awards

Programming Languages: Python, C, C++, JavaScript, Go, SQL

Systems: C++ for embedded systems, real-time signal processing, state machines

ML/AI: LLM integration, semantic search, vector embeddings, NLP, RAG

Cloud & Infrastructure: GCP (Vertex AI, Cloud Run), AWS (Lambda, DynamoDB), Docker, Redis

Tools: React, FastAPI, Git, Linux

Best Cybersecurity Hack, MHack16 (350+ participants) | Dec 2023

Native Languages: English, Polish